Applied Performance Task, Expectations and Rubric

The Task

This lesson looks at the Ottawa River Watershed in particular, using MFworks. It deals with building masks, created slope and shaded relief maps from DEM (Digital Elevation Model), composite overlays and data analysis. Students begin by examining various land use patterns, as they exist in the Ottawa River Watershed. Land use patterns include agriculture, urbanization, soils, and vegetation patterns as they relate to water quality. Students will research how different land use patterns affects water quality and in groups prepare for a debate focused on the improvement of water quality. The debate should focus on suggesting ways to prepare for the future, to insure a high water quality for future generations. Students are instructed to include supporting visuals, such as maps, graphs, charts, diagrams, or pictures, in the debate.

Expectations

This task gives students the opportunity to demonstrate achievement of the following selected expectations from three strands: Geographic Foundations: Space and Systems; Understanding and Managing Change; and Methods of Geographic Inquiry.

1. demonstrate an understanding of the challenges associated with achieving resource sustainability and explain the implications of meeting or not meeting those challenges for future resource use in Canada;
2. create and implement a plan to address a local environmental concern;
3. synthesize information on changes in the geography of Canada, such as changes in land use and urban patterns, as well as resource depletion, in order to plan for the future;
4. predict the consequences of human activities (e.g., agriculture, recreation) on natural systems (e.g., soil depletion, climate change);
5. use communication skills (e.g., letter writing, debating, consensus building) effectively to promote environmental awareness;
6. provide evidence to support conclusions and opinions;
7. create and use effectively photographs, charts, graphs, models, and diagrams.

Prior Knowledge and Skills

To complete this task, students are expected to be able to demonstrate the following:

- know ways in which rivers are important to our lives and our culture;
- know about runoff, drainage basins, drainage divide, river flow, etc.;
- know about major Canadian water bodies;
- know about the hydrologic cycle;
- have an idea of water as a resource, Canada's supply of fresh water and the demands for water resources;
- have an idea of how rivers and water basins get contaminated;
- ability to read, analyse, and interpret satellite imagery and graphs;
- ability to use a variety of tools and technologies for research;
- select and use appropriate methods for displaying geographic data;
- use geographic data to support conclusions and opinions;
- communicate the results of geographic inquiries using appropriate methods and technologies, and present viewpoints on issues affecting Canadians;
- use cartographic conventions correctly when constructing maps (e.g., scale, legend, direction).

GEOMATICS
FOR HIGH SCHOOL CURRICULUM

Task Rubric – Future Water Quality Debate

Expectations	Criteria	Level 1 (50 – 59)	Level 2 (60 –69)	Level 3 (70 – 79)	Level 4 (80–100)
Knowledge / understanding					
1	• Demonstrate an understanding of the challenges associated with achieving resource sustainability, and explain the implications of meeting or not meeting those challenges for future resource use in Canada	• Demonstrates little understanding of the challenges associated with achieving resource sustainability	• Demonstrates some understanding of the challenges associated with achieving resource sustainability	• Demonstrates considerable understanding of the challenges associated with achieving resource sustainability	• Demonstrates a high degree of understanding of the challenges associated with achieving resource sustainability
Thinking / Inquiry					
3, 4	• Predict the consequences of human activities on natural systems to plan for the future	• Fails to predict the consequences of human activities on natural systems to plan for the future	• Shows some consequences of human activities on natural systems to plan for the future	• Shows considerable consequences of human activities on natural systems to plan for the future	• Shows a high degree of consequences of human activities on natural systems to plan for the future
Communication					
5, 6, 7	• Use visuals and collected research to support conclusions and opinions in debate	• Uses visuals and collected research to support conclusions and opinions in debate to a limited degree	• Uses visuals and collected research to support conclusions and opinions in debate to some degree	• Uses visuals and collected research to support conclusions and opinions in debate to a considerable degree	• Uses visuals and collected research to support conclusions and opinions in debate to a high degree
Application					
2	• Create and implement a plan to address a local environmental concern;	• Fails to create and implement a plan	• Creates and implements of a plan to some degree	• Creates and implements a good plan	• Creates and implements a good and very efficient plan